

A Comparator Gap for Distributed CVaR Decisions

Q Learning-Augmented Algorithms: Guarantees, Construction Mechanisms, and System-Level Implications [arXiv:2609.04787](#)

P Distributed risk-averse optimization via CVaR [arXiv:2609.04460](#)

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Abstract

Wang et al. optimise an average of local conditional value-at-risk (CVaR) objectives, whereas deployment may incur the CVaR of the sum of the same losses. We identify the comparator term needed to transfer their finite-time bound to that aggregate objective and give a two-agent example in which the term is actual decision loss even at an exact local optimum. A fresh holdout sample supports a selective aggregate-risk certificate, subject to a declared observation interface and a safe fallback. Existing work has already established that the order of risk aggregation can change decisions; our result is a precise accounting for this particular optimisation guarantee and deployment interface.

1 The two source results and the question

Zhao et al. survey learning-augmented algorithms and explain why a component guarantee needs an explicit downstream benchmark relation, conditional interface guarantees, and cost accounting before it becomes a system guarantee [4]. Their priced quantity is a predictor invocation. Wang et al. study a different primitive: agents query sampled local losses and communicate decisions while optimising the average of their local CVaRs [3]. Their finite-time theorem bounds the *expected* objective gap of a weighted ergodic output; their projection makes its membership in the decision domain exact. Neither fact certifies a realised output against a separate aggregate-risk budget. These are statements of Q and P, respectively.

We consider only the identity deployment interface: the decision produced by P is used for the same agents' losses. Let the common tail level be $\alpha \in (0, 1)$, let $\mathcal{X}$ be P's decision set, and write

$$C_i(x) = \text{CVaR}_\alpha(J_i(x, \xi_i)), \quad C(x) = \frac{1}{m} \sum_{i=1}^m C_i(x),$$

$$S(x, \xi) = \sum_{i=1}^m J_i(x, \xi_i), \quad A(x) = \text{CVaR}_\alpha(S(x, \xi)).$$

Here ξ has the joint deployment law and each marginal agrees with the corresponding local evaluation law. This joint law is an additional specification: P's local-oracle model gives the marginals, but no paired aggregate-loss oracle. Set $C^* = \min_{x \in \mathcal{X}} C(x)$ and $A^* = \min_{x \in \mathcal{X}} A(x)$.

For P's weighted ergodic output $X = \hat{x}_T$, assume its theorem's conditions and denote a deterministic upper bound on its right-hand side by B_T . Thus

$$X \in \mathcal{X} \quad \text{pathwise,} \quad \mathbb{E}[C(X) - C^*] \leq B_T.$$

The expectation is over P's optimisation samples and algorithmic randomness. This paper adds no new finite-time estimate for B_T .

2 The aggregate comparator

Proposition 1 (Benchmark transfer). *Under the preceding common-tail and matched-loss assumptions, define*

$$D = mC^* - A^*.$$

Then $D \geq 0$ and

$$\mathbb{E}[A(X) - A^*] \leq mB_T + D.$$

Proof. CVaR subadditivity, also noted in P for a common tail level, gives $A(x) \leq mC(x)$ for every x . Minimising both sides gives $A^* \leq mC^*$. At X , add and subtract mC^* , then take expectations and apply P's bound. $\square$

The term D compares *different optima*. P's local suboptimality bound does not control it; P's appendix identifies the pointwise diversification price but does not supply this optimiser-level comparison. The proposition is an additive bridge of the kind Q's composition discussion requires, without yielding Q's competitive ratio for another downstream policy.

Proposition 2 (The gap can be attained). *Let $U > 0$, $m = 2$, $\alpha = 1/2$, $\mathcal{X} = [-1, 1]$, and let Z be a shared fair Bernoulli variable. For*

$$J_1(x, Z) = UZ, \quad J_2(x, Z) = U((1-x)Z + x(1-Z)),$$

the exact minimiser of C is $x_P = 1/2$, while the minimiser of A is $x_A = 1$. Moreover,

$$A(x_P) - A^* = D = U/2.$$

Proof. The local and aggregate CVaRs are

$$\begin{aligned} C_1(x) &= U, & C_2(x) &= U \max\{1-x, x\}, \\ A(x) &= U \max\{2-x, x\} = U(2-x) & \text{for } x \in [-1, 1]. \end{aligned}$$

Hence $2C^* = 3U/2$, $A^* = U$, and the displayed gap follows. Both losses are bounded and affine in x , as required for P's loss class. $\square$

This example shows actual aggregate decision loss, beyond looseness of a certificate at a fixed point. It does not assert that D equals the decision loss in general.

3 A conditional deployment certificate

The aggregate benchmark comparison is an expectation bound. To assess a realised X , freeze it and draw n_i fresh validation losses from each marginal law, independently of P's

optimisation transcript. Suppose $|J_i| \leq U_i$, and let $\widehat{C}_i(X)$ be the empirical CVaR of this batch. For $\zeta \in (0, 1)$, put

$$\epsilon_i = \frac{2U_i}{\alpha} \sqrt{\frac{\log(2m/\zeta)}{2n_i}}, \quad R = \sum_{i=1}^m (\widehat{C}_i(X) + \epsilon_i).$$

Conditional on the optimisation transcript, the DKW inequality and P's CVaR stability lemma give, with validation probability at least $1 - \zeta$, simultaneous errors $|\widehat{C}_i(X) - C_i(X)| \leq \epsilon_i$. Consequently, on this event,

$$A(X) \leq \sum_i C_i(X) \leq R \leq mC(X) + 2 \sum_i \epsilon_i.$$

Freshness matters because X was chosen using the optimisation samples. CVaR subadditivity requires no independence across agents in the deployment law.

For a fixed risk budget b , accepting X only if $R \leq b$ therefore gives

$$\Pr\{\text{accept } X \text{ and } A(X) > b\} \leq \zeta.$$

On the validation event, the risk statement also implies $\Pr_{\xi}\{S(X, \xi) > R \mid X, \text{validation}\} \leq \alpha$ for a fresh deployment draw. The confidence probability over the audit and the tail probability over future loss are different.

Proposition 3 (Selective deployment). *Suppose a known $x_0 \in \mathcal{X}$ satisfies $A(x_0) \leq b$, and suppose a known margin $mC^* \leq b - h$ holds for $h > 0$. If $2 \sum_i \epsilon_i \leq h/2$, deploy X when $R \leq b$ and x_0 otherwise. Write $\Delta = C(x_0) - C^*$. Then*

$$\begin{aligned} \Pr\{\text{reject } X\} &\leq \zeta + \frac{2mB_T}{h}, \\ \Pr\{A(X_{\text{deploy}}) > b\} &\leq \zeta, \\ \mathbb{E}[C(X_{\text{deploy}}) - C^*] &\leq B_T + \Delta \left(\zeta + \frac{2mB_T}{h} \right). \end{aligned}$$

Proof. On the simultaneous validation event, $R \leq mC(X) + 2 \sum_i \epsilon_i$. The margin and precision conditions make $R \leq b$ whenever $C(X) - C^* \leq h/(2m)$. Markov's inequality and P's expected gap bound prove the first claim. The fallback is known safe, so only a false acceptance on the failed validation event can breach b . Finally, the deployed local gap is at most the gap at X plus Δ on rejection; take expectations. $\square$

The margin and safe fallback are additional premises, not outputs of P. The proposition concerns the same action and same losses, rather than an arbitrary downstream policy.

4 Observation and cost limits

Marginal validation can be arbitrarily conservative. If exactly one of m agents incurs U in each equally likely outcome and $\alpha = 1/m$, then $\sum_i C_i = mU$ but $A = U$. Moreover, two agents with losses (UZ, UZ) or $(UZ, U(1 - Z))$, for a fair Bernoulli Z , have the same local marginals but aggregate CVaRs $2U$ and U at $\alpha = 1/2$. With only separately generated marginal oracle samples, their transcripts cannot reveal which coupling will occur in deployment. This last statement assumes no synchronised cross-agent outcomes or other dependence information.

A fresh paired audit of the actual joint loss at frozen X can instead estimate $A(X)$ directly. With n independent paired vectors, $U_{\Sigma} = \sum_i U_i$, and empirical aggregate CVaR $\widehat{A}(X)$, set

$$\epsilon_J = \frac{2U_{\Sigma}}{\alpha} \sqrt{\frac{\log(2/\zeta)}{2n}}, \quad R_J = \widehat{A}(X) + \epsilon_J.$$

The same DKW argument gives $A(X) \leq R_J \leq A(X) + 2\epsilon_J$ with confidence $1 - \zeta$. This removes the local-sum slack at the selected decision, but does not identify A^* or remove D from Proposition 1. Paired sensing and communication are extra interface requirements.

If an offline local evaluation costs κ_L and a paired validation vector costs κ_J , the declared acquisition charge is $\kappa_L \sum_{k < T, i} s_k^i + \kappa_J n$. A local audit instead charges its $\sum_i n_i$ evaluations. These are accounting conventions inspired by Q, whose own charge is for predictor calls. They exclude forming X , audit communication, and any loss from physically playing a queried action; an application must price those as well. The displayed risk bounds are not cost-augmented competitive ratios.

For a deterministic call schedule and an all-in budget B in the same units, the risk-only acceptance threshold must subtract the declared acquisition charge:

$$b = B - \kappa_L \sum_{k < T, i} s_k^i - \kappa_J n.$$

Any additional implementation charge must also be subtracted before the risk filter is applied.

5 Related work and interpretation

The distinction between risk of an aggregate and aggregation of individual risks is already published. Almen and Dentcheva [1] compare those evaluation orders in a distributed system and report different decisions and less conservative aggregate risk evaluation. Thus the main aggregation insight and the possibility of changed decisions are prior work. P itself notes the pointwise diversification price [3]. The contribution here is the exact additive comparator term for P 's finite-time guarantee, together with its tight two-agent instance and the assumptions needed to audit its output in Q's interface language.

CVaR confidence bounds are also established. Liang and Luo [2] study risk-measure confidence bounds, including CVaR, and improve on empirical-risk-plus-radius constructions. The DKW holdout radius used above is a direct combination of standard concentration with P's stability lemma, not a new estimation method. The examples about dependence are likewise illustrations of a known risk-aggregation obstruction, specialised to this oracle interface.

6 Limitations and open questions

No finite-sample filter that accepts a candidate on an all-zero safe transcript can guarantee exact distribution-free risk feasibility for that candidate across the stated loss class. The ledger gives a bounded, convex, Lipschitz single-agent example with a safe fallback: the candidate loss is $J(x, Z) = \max\{0, x\}Z$, with fallback $x_0 = 0$ and candidate $x = 1$. An all-zero law makes the candidate safe; under a law where $Z = U$ with probability $p \in (\alpha b/U, \alpha)$, its CVaR is $Up/\alpha > b$. Every finite all-zero transcript occurs with positive probability under both laws. A rule accepting the candidate on that transcript can therefore falsely accept it. Exact membership of X by P's projection remains a separate fact.

The analysis also assumes a common tail level, matched local and deployment losses, fresh validation, and either a safe fallback or abstention. A paired audit needs access to the joint deployment law. The cost expression omits implementation-specific communication and physical-play losses. No bound on D , no aggregate-aware training method, and no competitive ratio for a separate downstream policy follow from the present material.

A concrete next question is whether a specified application can pay for local training, joint observation, validation and fallback while also bounding D tightly enough to improve on its

known safe baseline. Q's downstream sensitivity and benchmark conditions would still need proof if the deployed policy differs from the decision considered here. The current results do not establish such an improvement.

References

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